

RAJ KUMAR TIWARI

Full Stack Developer | AI & LLM Integration

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[GitHub](#) | [LinkedIn](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Full Stack Developer with **2.5+ years of experience** designing and shipping responsive, high-performance web applications using **React.js, Next.js, JavaScript, TypeScript, Redux**, and **Material UI**, backed by **Node.js, Express.js**, and **MongoDB**. Proven record of turning complex requirements into scalable architecture and measurable gains in performance, with intermediate expertise in **Data Structures & Algorithms** and **C++**, and strong proficiency in data persistence with **MongoDB**. Hands-on with **Python, FastAPI, WebSockets, Redis, Docker**, and **Kafka**, and experienced in delivering production AI features through **Retrieval-Augmented Generation (RAG)**, **LangChain**, and **OpenAI** integration. Seeking to apply full stack and applied-AI engineering skills to build reliable, user-focused products at scale.

TECHNICAL SKILLS

Frontend: React.js, Next.js, JavaScript (ES6+), TypeScript, Redux, Material UI, HTML5, CSS3, Responsive UI

Backend: Node.js, Express.js, Python, FastAPI, REST APIs, WebSockets, Kafka

Databases: MongoDB, Redis

AI & LLM: Retrieval-Augmented Generation (RAG), LangChain, OpenAI APIs, FAISS Vector Database

Cloud & DevOps: Docker, Git

Core Competencies: Data Structures & Algorithms, C++, Object-Oriented Programming, System Design Fundamentals, Agile / Scrum

PROFESSIONAL EXPERIENCE

Software Engineer — Sense & Respond Software LLC

Jan 2024 - Apr 2026

Noida, India

- Built responsive, high-performance user interfaces with **React.js, Redux, TypeScript, JavaScript**, and **Material UI**, prioritizing scalability, accessibility, and a consistent user experience across modules.
- Integrated **REST APIs** with **Redux** for centralized state management, streamlining data flow across the application and reducing redundant network calls.
- Implemented **Web Workers** to offload computationally intensive tasks, improving UI responsiveness and **reducing page load time by approximately 30%** by preventing main-thread blocking.
- Introduced **Redis caching** for frequently accessed data, cutting database queries and **improving API response time by approximately 40%** while improving overall application scalability.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline with **LangChain** and **OpenAI** that grounded model responses in retrieved contextual documents, **reducing AI hallucinations by approximately 50%**.
- Developed and integrated **OpenAI-powered LLM APIs** into full stack applications, enabling intelligent features such as real-time responses and enhanced user interaction.
- Developed server-side logic and **REST APIs** using **Node.js** and **Express.js** with **MongoDB** integration, supporting end-to-end feature delivery from database to interface.
- Collaborated with product managers, designers, and backend engineers in an **Agile** environment; participated in **code reviews** and improved maintainability through reusable, **component-based architecture**.

Software Engineer Intern — Sense & Respond Software LLC

Sep 2023 - Dec 2023

Noida, India

- Built **reusable UI components** using **React.js, TypeScript, JavaScript, Redux**, and **Material UI**, contributing to a scalable and consistent frontend architecture.
- Gained practical exposure to backend fundamentals including **API handling**, data flow, and frontend-backend integration, strengthening full stack development capability.

PROJECTS

AI Agent with Retrieval-Augmented Generation (RAG)

LangChain, FAISS, Llama 3.1

- Developed an intelligent **AI agent** with **LangChain** that draws on multiple data sources and tools to deliver context-aware, accurate responses.
- Built a **RAG pipeline** that loads web-based documents, splits text into chunks, generates **embeddings**, and stores them in a **FAISS vector database** for efficient semantic retrieval.

- Integrated external knowledge sources including the **Wikipedia** and **arXiv APIs** to broaden topical coverage.
- Created a **custom retriever tool** and enabled dynamic tool selection through a **tool-calling agent architecture**, with **ChatOllama (Llama 3.1)** as the core LLM.
- Implemented an **Agent Executor** to process user queries and fetch relevant information from multiple sources in real time.

Review Portal

React.js, TypeScript, Node.js, OpenAI

- Designed, developed, and deployed a **full stack review management platform** consisting of a user portal and an admin portal, including both frontend and backend implementation.
- **User Portal:** Built with **React.js, TypeScript, Material UI**, and **Redux**, allowing users to generate reviews with **OpenAI**, submit them, and upload screenshots as proof of submission.
- **Admin Portal:** Built an administrative dashboard to moderate submissions, verify uploaded screenshots, and approve or reject reviews.

Product Inventory Management System

FastAPI, PostgreSQL, React.js, Docker

- Developed **RESTful APIs** with **FastAPI** for product, customer, and order management, backed by relational data models using **SQLAlchemy ORM** and **PostgreSQL**.
- Implemented **Pydantic validation**, inventory tracking, stock validation, and automated order processing workflows with **transaction management** and structured error handling.
- Containerized backend services with **Docker** for consistent, reproducible deployments across environments.
- Built a responsive **React.js** frontend integrated with backend APIs via **Axios**, delivering real-time **CRUD operations** through reusable form and table components.

Trading View Platform

Next.js, TypeScript

- Developing a modern trading platform in **Next.js** with a high-performance, scalable frontend focused on a seamless user experience.
- Implementing **dynamic UI components** and **real-time data visualization**, with planned extension into backend services and APIs.

CheckIn Mobile Application

Flutter, Dart

- Developed a **cross-platform mobile application** for check-in and check-out management, using a lightweight **local JSON-based database** for data handling and **offline support**.

EDUCATION

B.Tech in Computer Science and Engineering (AI & ML)

2020 - 2024

Galgotias University, Greater Noida, Uttar Pradesh, India